Assignment #1 Within-VR Performance Conditioned on Cognitive Function

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This analysis examined the relationship between participants' visuospatial cognitive ability and their task efficiency in a virtual reality (VR) cooking simulation. The Block Design score was used as an indicator of visuospatial reasoning, while total recipe completion time represented participants' cognitive and motor performance. Data from 13 participants who completed the "Simple Stew" VR task were included in the analysis.

Descriptive Analysis

Descriptive statistics summarized the range and distribution of participant scores and performance measures. The Block Design score had a mean of 39.31 (SD = 12.32), with scores ranging from 24 to 63. The Total Recipe Completion Time ranged from 198.27 to 580.02 seconds (M = 385.30, SD = 128.74). The variability in efficiency possibly reflects differences in spatial planning, working memory, or familiarity with VR-based interaction.

Table 1

Participant Total Completion Time

Participant ID	Total Completion Time (s)
C008	352.99
C009	452.49
C013	451.59
C016	198.27
C017	403.61
C021	580.02
C024	200.09
C025	430.63
C032	251.38
C033	476.88
C037	542.51
C040	261.57
C041	324.16

Correlation Between Cognitive and Performance Measures

A Spearman correlation analysis was conducted to explore the association between *Block Design Scores* and *Total Completion Times*. The correlation was negative and non-significant, $r(11) = -0.397$, $p = 0.1796$.

Although not statistically significant, this negative direction indicates that participants with higher visuospatial ability generally completed the VR cooking task more quickly than those with lower scores. This pattern supports the expectation that cognitive factors influence the speed and fluidity of goal-directed movements in immersive environments.

Linear Regression Analysis

A linear regression model examined the predictive relationship between cognitive performance and task duration. The resulting model was expressed as:

$$Y = -0.0357x + 52.830$$

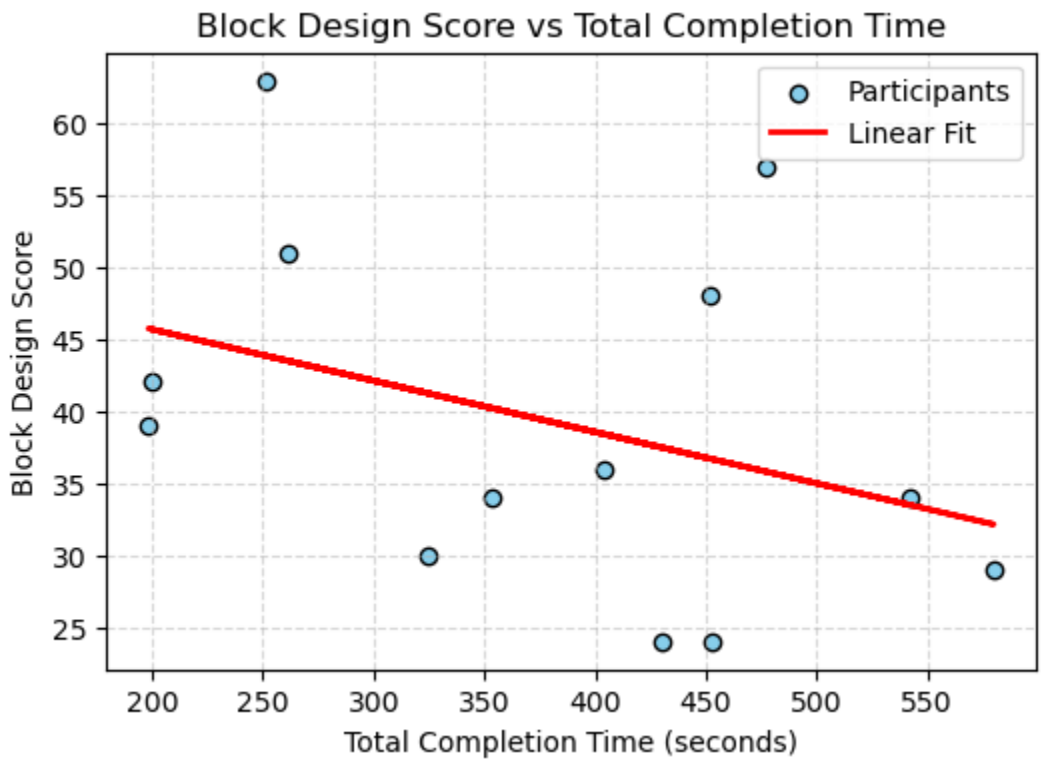
where x represents the total completion time (in seconds) and y predicts the Block Design score. The **negative slope ($\beta_1 = -0.0357$)** suggests a weak inverse relationship: shorter task durations were associated with higher predicted cognitive performance. The **intercept ($\beta_0 = 52.83$)** represents the expected cognitive score corresponding to minimal task time.

Visualization of the Relationship

Figure 1 depicts the relationship between Block Design Score and Total Completion Time, with individual participant data points and the linear fit line superimposed. The graph exhibits a mild downward trend, reflecting an inverse, though statistically non-significant, relationship between cognitive performance and motor efficiency.

Figure 1

Scatter plot showing the relationship between Block Design Score and Total Completion Time (s), with linear regression line (red) indicating the trend direction.



Proposed Quantitative Measure: Pacing Consistency Index (PCI)

The proposed quantitative measure for this investigation was the Pacing Consistency Index (PCI). PCI quantifies the regularity and consistency with which participants progress through sequential steps during a task. It builds on the coefficient of variation, which is the ratio of the standard deviation to the mean, which reflects variability in time-based measures such as step intervals (Six Sigma.us, 2024). A higher CV indicates greater dispersion of data points around the mean, suggesting higher relative variability. CV has been used in performance and rehabilitation research to assess stability and control. Matsuda et al. (2015) used stride-time CV

to evaluate gait variability among older adults. The authors found that higher CV values indicated poorer rhythmic stability and increased fall risk.

To adapt this concept for task pacing in VR performance data, the inverse monotonic transformation was applied to standardize and simplify interpretation. Higher values represent steadier, more rhythmic performance, while lower values denote irregular timing or disruptions in task flow.

$$PCI = \frac{1}{1 + CV}$$

where the coefficient of variation (CV) represents the ratio of the standard deviation of step intervals to their mean. This formulation standardizes the variability of pacing, producing a value between 0 and 1 that reflects the degree of consistency in task execution.

$$CV = \frac{SD \text{ of Step Intervals}}{Mean \text{ of Step Intervals}}$$

Table 2

PCI Interpretation

PCI Value	***Interpretation***
Closer to 1.0	Highly consistent pacing, steady rhythm and minimal variability between steps
Around 0.5	Alternating between steady and delayed responses
Closer to 0	Frequent pauses or disruptions in task flow

Table 3

Pacing Consistency Index (PCI) of the Participant

Participant ID	Mean Interval	SD Interval	Coefficient of Variation (CV)	Pacing Consistency Index (PCI)
C008	9.796	11.239	1.147	0.466
C009	11.836	9.688	0.819	0.550
C013	13.703	22.890	1.670	0.374
C016	5.696	6.860	1.204	0.454
C017	9.537	14.207	1.490	0.402
C021	13.300	21.887	1.646	0.378
C024	5.707	7.543	1.322	0.431
C025	11.084	16.201	1.462	0.406
C032	7.002	11.844	1.691	0.372
C033	11.480	14.430	1.257	0.443
C037	11.538	15.777	1.367	0.422
C040	7.536	11.661	1.547	0.393
C041	8.699	10.689	1.229	0.449

Hypothesis testing

To test the validity of the proposed PCI, we plan to use Spearman's correlation to examine its relationship with Block Design scores, a cognitive measure of visuospatial reasoning. We hypothesize that higher PCI values, indicating steadier and more consistent pacing, will be positively associated with higher cognitive performance.

Null Hypothesis (H_0):

There is no significant correlation between participants' Pacing Consistency Index (PCI) and their Block Design Scores. In other words, pacing stability during the task is not related to visuospatial reasoning ability.

Alternative Hypothesis (H_1):

There is a significant correlation between participants' Pacing Consistency Index (PCI) and their Block Design Scores. Specifically, it is expected that higher PCI values (greater consistency) will be associated with higher Block Design Scores, reflecting better cognitive–motor coordination.

Spearman Correlation Analysis and Hypothesis results

A Spearman correlation was conducted to examine the relationship between the Pacing Consistency Index and Block Design Scores. PCI served as the independent variable representing pacing stability, and Block Design Score served as the dependent variable reflecting cognitive performance. The correlation was negative but non-significant, $\rho(13) = -.41$, $p = .16$, indicating a weak inverse association between pacing consistency and cognitive performance. The null hypothesis (H_0), stating no significant relationship, could not be rejected. Further research with a larger and more diverse sample could help clarify whether this observed trend reflects a true relationship or random variation

Linear Regression Analysis A Spearman rank-order correlation was conducted to examine the relationship between the Pacing Consistency Index and Block Design Scores. PCI served as the independent variable representing pacing stability, and Block Design Score served as the dependent variable reflecting cognitive performance. The correlation was negative but non-significant, $\rho(13) = -.41$, $p = .16$, indicating a weak inverse association between pacing consistency and cognitive performance. The null hypothesis (H_0), stating no significant relationship, could not be rejected. Further research with a larger and more diverse sample could help clarify whether this observed trend reflects a true relationship or random variation.

A simple linear regression was conducted to examine the relationship between the Pacing Consistency Index (PCI) and Block Design scores. The resulting model was:

$$y = -113.07x + 87.49$$

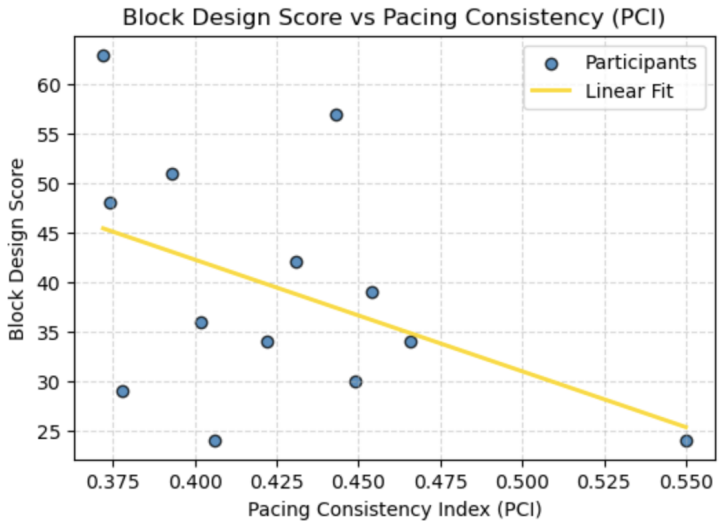
where x represents PCI and y represents the predicted cognitive score. The negative slope ($\beta = -113.07$) indicates a weak inverse association: participants with more consistent pacing tended to have slightly lower predicted Block Design scores, although the relationship was not statistically significant. However, the limited sample size and small variance in PCI values constrain the model's predictive strength.

Visualization of PCI–Block Design Relationship

As shown in Figure 2, the scatterplot illustrates the relationship between participants' Pacing Consistency Index (PCI) and Block Design Score. The yellow regression line depicts a weak negative association between PCI and Block Design Scores, consistent with the direction and strength observed in the correlation analysis.

Figure 2

Scatterplot of Pacing Consistency Index (PCI) & Block Design Score with Linear Regression



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Appendix

ChatGPT Log

Student	***URL***
Sri ramya Panja	hhttps://chatgpt.com/share/68f44f47-9c8c-800d-8522-b191840fafdf
Lavanya Ruthala	https://chatgpt.com/share/68f51e43-8ac0-8008-b474-b103a964a965
Garen Schroder	https://chatgpt.com/share/68f456ef-3bc0-8008-9b6c-6317404c93ae